KAIZEN MATHS

Lesson starter

Proof by Induction

Divisibility by Induction

Name _____

S1

Prove each divisibility result by mathematical induction.

Prove by induction that $7^n - 1$ is divisible by 6 for $n \geq 1$.

S2

Prove each divisibility result by mathematical induction.

Prove by induction that $9^n - 4^n$ is divisible by 5 for $n \geq 1$.

S3

Prove each polynomial divisibility result by mathematical induction.

Prove by induction that $n^2 + n$ is divisible by 2 for $n \geq 1$.

S4

Prove each divisibility result by mathematical induction.

Prove by induction that $6^n - 1$ is divisible by 5 for $n \geq 1$.

Show your working

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Show your working

KAIZEN MATHS

Exit ticket

Proof by Induction

Divisibility by Induction

Name _____

EA1

Prove each divisibility result by mathematical induction.

Prove by induction that $8^n - 5^n$ is divisible by 3 for $n \geq 1$.

EA2

Prove each polynomial divisibility result by mathematical induction.

Prove by induction that $n^3 + 2n$ is divisible by 3 for $n \geq 1$.

Show your working

KAIZEN MATHS

Exit ticket

Proof by Induction

Divisibility by Induction

Name _____

EB1

Prove each divisibility result by mathematical induction.

Prove by induction that $5^n - 1$ is divisible by 4 for $n \geq 1$.

Show your working

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Exit ticket

Proof by Induction

Divisibility by Induction

Name _____

EA1

Prove each divisibility result by mathematical induction.

Prove by induction that $8^n - 5^n$ is divisible by 3 for $n \geq 1$.

EA2

Prove each polynomial divisibility result by mathematical induction.

Prove by induction that $n^3 + 2n$ is divisible by 3 for $n \geq 1$.

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Show your working

KAIZEN MATHS

Exit ticket

Proof by Induction

Divisibility by Induction

Name _____

EB1

Prove each divisibility result by mathematical induction.

Prove by induction that $5^n - 1$ is divisible by 4 for $n \geq 1$.

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Show your working

Teacher answers & working

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S1

Prove each divisibility result by mathematical induction.

Prove by induction that $7^n - 1$ is divisible by 6 for $n \geq 1$.

Answer:

Proved by induction.

Working:

Base case n = 1: $7^1 - 1 = 6$, which is divisible by 6.

Assume true for n = k: $7^k - 1 = 6m$ for some integer m.

For $n = k + 1$: $7^{k+1} - 1 = 7(7^k) - 1$.

$= 7(7^k - 1) + (7 - 1)$

$= 7(6m) + (6)$

$= (6)(7m + 1)$.

Since $7m + 1$ is an integer, $7^{k+1} - 1$ is divisible by 6.

The base case has been shown to be true. The induction step has shown that if the statement is true for n = k, then it is true for n = k + 1. Therefore, by mathematical induction, the statement is true for all integers $n \geq 1$.

S2

Prove each divisibility result by mathematical induction.

Prove by induction that $9^n - 4^n$ is divisible by 5 for $n \geq 1$.

Answer:

Proved by induction.

Working:

Base case n = 1: 9 − 4 = 5, which is divisible by 5.

Assume true for n = k: $9^k - 4^k = 5m$ for some integer m.

For $n = k + 1$: $9^{k+1} - 4^{k+1} = 9(9^k) - 4(4^k)$.

$= 9(9^k - 4^k) + (9 - 4)4^k$

$= 9(5m) + 54^k$

$= 5(9m + 4^k)$.

Since $9m + 4^k$ is an integer, the expression is divisible by 5.

The base case has been shown to be true. The induction step has shown that if the statement is true for n = k, then it is true for n = k + 1. Therefore, by mathematical induction, the statement is true for all integers $n \geq 1$.

Teacher answers & working

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S3

Prove each polynomial divisibility result by mathematical induction.

Prove by induction that $n^2 + n$ is divisible by 2 for $n \geq 1$.

Answer:

Proved by induction.

Working:

Base case n = 1: $1^2 + 1 = 2$, so the result is true.

Assume true for n = k: $k^2 + k = 2m$ for some integer m.

For $n = k + 1$: $(k + 1)^2 + (k + 1)$

$= k^2 + 2k + 1 + k + 1$

$= (k^2 + k) + 2k + 2$

$= 2m + 2(k + 1) = 2(m + k + 1)$.

This is divisible by 2.

The base case has been shown to be true. The induction step has shown that if the statement is true for n = k, then it is true for n = k + 1. Therefore, by mathematical induction, the statement is true for all integers $n \geq 1$.

S4

Prove each divisibility result by mathematical induction.

Prove by induction that $6^n - 1$ is divisible by 5 for $n \geq 1$.

Answer:

Proved by induction.

Working:

Base case n = 1: $6^1 - 1 = 5$, which is divisible by 5.

Assume true for n = k: $6^k - 1 = 5m$ for some integer m.

For $n = k + 1$: $6^{k+1} - 1 = 6(6^k) - 1$.

$= 6(6^k - 1) + (6 - 1)$

$= 6(5m) + (5)$

$= (5)(6m + 1)$.

Since $6m + 1$ is an integer, $6^{k+1} - 1$ is divisible by 5.

The base case has been shown to be true. The induction step has shown that if the statement is true for n = k, then it is true for n = k + 1. Therefore, by mathematical induction, the statement is true for all integers $n \geq 1$.

Teacher answers & working

Proof by Induction · Divisibility by Induction

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EA1

Prove each divisibility result by mathematical induction.

Prove by induction that $8^n - 5^n$ is divisible by 3 for $n \geq 1$.

Answer:

Proved by induction.

Working:

Base case n = 1: 8 − 5 = 3, which is divisible by 3.

Assume true for n = k: $8^k - 5^k = 3m$ for some integer m.

For $n = k + 1$: $8^{k+1} - 5^{k+1} = 8(8^k) - 5(5^k)$.

$= 8(8^k - 5^k) + (8 - 5)5^k$

$= 8(3m) + 35^k$

$= 3(8m + 5^k)$.

Since $8m + 5^k$ is an integer, the expression is divisible by 3.

The base case has been shown to be true. The induction step has shown that if the statement is true for n = k, then it is true for n = k + 1. Therefore, by mathematical induction, the statement is true for all integers $n \geq 1$.

EA2

Prove each polynomial divisibility result by mathematical induction.

Prove by induction that $n^3 + 2n$ is divisible by 3 for $n \geq 1$.

Answer:

Proved by induction.

Working:

Base case n = 1: $1^3 + 2(1) = 3$, so the result is true.

Assume true for n = k: $k^3 + 2k = 3m$ for some integer m.

For $n = k + 1$: $(k + 1)^3 + 2(k + 1)$

$= k^3 + 3k^2 + 3k + 1 + 2k + 2$

$= (k^3 + 2k) + 3k^2 + 3k + 3$

$= 3m + 3(k^2 + k + 1) = 3(m + k^2 + k + 1)$.

This is divisible by 3.

The base case has been shown to be true. The induction step has shown that if the statement is true for n = k, then it is true for n = k + 1. Therefore, by mathematical induction, the statement is true for all integers $n \geq 1$.

Teacher answers & working

Proof by Induction · Divisibility by Induction

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EB1

Prove each divisibility result by mathematical induction.

Prove by induction that $5^n - 1$ is divisible by 4 for $n \geq 1$.

Answer:

Proved by induction.

Working:

Base case n = 1: $5^1 - 1 = 4$, which is divisible by 4.

Assume true for n = k: $5^k - 1 = 4m$ for some integer m.

For $n = k + 1$: $5^{k+1} - 1 = 5(5^k) - 1$.

$= 5(5^k - 1) + (5 - 1)$

$= 5(4m) + (4)$

$= (4)(5m + 1)$.

Since 5m + 1 is an integer, $5^{k+1} - 1$ is divisible by 4.

The base case has been shown to be true. The induction step has shown that if the statement is true for n = k, then it is true for n = k + 1. Therefore, by mathematical induction, the statement is true for all integers $n \geq 1$.